

PART 1: AI CONCEPTS

What is Artificial Intelligence?

1)Artificial Intelligence (AI) is a branch of computer science that focuses on creating machines and software capable of performing tasks that normally require human intelligence. These tasks include learning from experience, understanding language, recognizing images and speech, solving problems, making decisions, and adapting to new situations.

AI systems analyze data, identify patterns, and use algorithms to make predictions or take actions. Unlike traditional programs that follow fixed instructions, AI can improve its performance by learning from data and experience.

2)Difference between AI, Machine Learning, Generative AI, and Agentic AI

AI (Artificial Intelligence): The broad field of making machines perform tasks that require human intelligence.

Machine Learning (ML): A subset of AI that learns from data and improves automatically.

Generative AI: A type of AI that creates new content such as text, images, audio, or code.

Agentic AI: An advanced AI that can plan, make decisions, and perform tasks autonomously to achieve goals.

Example:

AI → Recommends movies.

ML → Learns your favorite movie genres.

Generative AI → Creates a movie review.

Agentic AI → Searches, compares, and books movie tickets for you.

3)Explain any three open-source AI models available on Hugging Face.

Three Open-Source AI Models Available on Hugging Face

1. **Llama 3** – An open-source language model by Meta used for text generation, question answering, and coding tasks.
2. **Mistral 7B** – A lightweight and efficient AI model known for strong performance in chatbots and language-processing tasks.
3. **Falcon** – An open-source language model developed by the Technology Innovation Institute, used for text understanding and content generation.

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4)What is the difference between traditional programming and AI-assisted development?

Difference Between Traditional Programming and AI-Assisted Development

Traditional Programming	AI-Assisted Development
Developers write all code manually.	AI helps generate and suggest code.
Follows fixed rules written by the programmer.	Learns from patterns and provides intelligent suggestions.
More time-consuming for repetitive tasks.	Faster development through automation.
Debugging is done manually.	AI can help detect and fix errors.
Output depends entirely on written instructions.	AI assists in creating solutions based on context.

Traditional Programming: Developers write all the rules and instructions manually, and the computer follows them exactly.

AI-Assisted Development: AI tools help developers by generating code, suggesting solutions, finding errors, and automating repetitive tasks.

5)Explain AI hallucinations with one example give me some matter

AI Hallucinations

AI hallucinations occur when an AI system generates information that appears correct and convincing but is actually false, inaccurate, or made up. This happens because AI predicts responses based on patterns in data rather than truly understanding facts.

Example:

If a user asks an AI for a research paper on a specific topic, the AI might provide a paper title, author names, and publication details that do not actually exist. Although the information sounds believable, it is incorrect and is considered an AI hallucination.

In short: AI hallucinations are false or misleading outputs generated by AI that are presented as if they were true.

AI-generated image :

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ChatGPT

Share

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Create an image representing Artificial Intelligence and Machine Learning.

ChatGPT



Here is an image representing Artificial Intelligence and Machine Learning.



Message ChatGPT



ChatGPT can make mistakes. Check important info.

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